

Blood Donation Management System

Group Members

Shumaila Zafar
Zobia Shaheen
Rameen Fatima

Program

BS Artificial Intelligence 3rd Regular

Course Instructor

Dr. Muhammad Ilyas

Introduction

This report explains the normalization process of the Blood Donation Management System database. The purpose of normalization is to organize the database efficiently, reduce redundancy, and improve consistency. The database is normalized step by step from Unnormalized Form (UNF) to First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF).

Unnormalized Form (UNF)

Initially, all information is stored in a single table. This causes redundancy and makes the database difficult to maintain.

DonorID	DonorName	BloodGroup	HospitalName	HospitalCity	BloodUnitID	Quantity	RequestID	PatientName
D01	Ali	A+	City Hospital	Lahore	BU01	2	R01	Ahmed
D01	Ali	A+	City Hospital	Lahore	BU02	1	R02	Hamza
D02	Sara	B+	Mayo Hospital	Lahore	BU03	3	R03	Fatima

Problems in UNF:

- Data redundancy exists.
- Donor and hospital information repeats multiple times.
- Update anomalies may occur.
- Deletion of one row can remove important information.

First Normal Form (1NF)

In First Normal Form, repeating groups are removed and each field contains only atomic values.

Steps Performed:

1. Removed repeating groups.
2. Ensured that each column contains a single value.
3. Converted all entries into atomic form.

DonorID	DonorName	BloodGroup	HospitalName	HospitalCity	BloodUnitID	Quantity	RequestID	PatientName
D01	Ali	A+	City Hospital	Lahore	BU01	2	R01	Ahmed
D01	Ali	A+	City Hospital	Lahore	BU02	1	R02	Hamza
D02	Sara	B+	Mayo Hospital	Lahore	BU03	3	R03	Fatima

Second Normal Form (2NF)

In Second Normal Form, partial dependencies are removed by separating related data into different tables.

Steps Performed:

1. Separated donor information into a separate table.
2. Separated hospital information into another table.
3. Created separate blood unit and request tables.
4. Removed repeated donor and hospital details.

DONOR Table

DonorID	DonorName	BloodGroup
D01	Ali	A+
D02	Sara	B+

HOSPITAL Table

HospitalID	HospitalName	HospitalCity
H01	City Hospital	Lahore
H02	Mayo Hospital	Lahore

BLOOD_UNIT Table

BloodUnitID	DonorID	Quantity
BU01	D01	2
BU02	D01	1
BU03	D02	3

Third Normal Form (3NF)

In Third Normal Form, transitive dependencies are removed. Each non-key attribute depends only on the primary key.

Steps Performed:

1. Created a separate BLOOD_GROUP table.
2. Connected donor table using foreign keys.
3. Removed transitive dependencies.
4. Improved database consistency and integrity.

BLOOD_GROUP Table

BloodGroupID	GroupName
BG01	A+
BG02	B+

DONOR Table

DonorID	DonorName	BloodGroupID
D01	Ali	BG01
D02	Sara	BG02

Conclusion

The Blood Donation Management System database was successfully normalized up to Third Normal Form (3NF). Normalization reduced redundancy, improved consistency, and made the database more efficient and maintainable.